

CareerForge & ubix

AUDIT STATUS: ACTION REQUIRED

Date:September 28, 2026

Environment: Next.js 14.2 / Node.js 24 / Windows

Audit Type: Codebase, Runtime & Architecture Audit

Target Scope: AI Assistant, Voice Engine, Auth, Security, UI

1. Executive Summary

A comprehensive technical, security, and runtime audit was conducted across the CareerForge / ubix platform. While the platform features high-quality visual components and rich multi-provider AI cascade scaffolding, several **critical architectural flaws** currently prevent core features—specifically the **AI Assistant**, **Voice Onboarding**, and **Continuous Speech Recognition**—from functioning as intended.

Subsystem	Status	Primary Root Cause & Impact
AI Assistant API <code>/api/assistant/chat</code>	CRITICAL	Client stores login in <code>localStorage</code> but fails to sync server-side <code>cf_session</code> cookie during OAuth/Phone sign-in or page reloads. Server rejects with <code>401 Unauthorized</code> , printing " <code>⚠️ Please sign in to interact with the assistant.</code> "
Voice Assistant & Dictation <code>GlobalVoiceDictator.tsx</code>	CRITICAL	Landing page voice onboarding calls <code>/api/assistant/chat</code> unauthenticated, resulting in 401 rejections and total audio silence. In chat, silence auto-send requires <code>isFinal===true</code> , hanging indefinitely on interim streams.
Design System Tokens <code>scripts/lint_design_tokens.mjs</code>	FAILING	88 hardcoded raw hex colors in <code>LandingPage.tsx</code> and <code>UboxCareerGraph.tsx</code> violate the zero-seam design token rules, breaking dark/high-contrast adaptability and causing <code>npm run test</code> to fail.
Test Suite & Services <code>v3_comprehensive_suite.test.mjs</code>	FAILING	Unit tests hardcode direct HTTP calls to <code>http://127.0.0.1:8000</code> (FastAPI Python backend), throwing <code>ECONNREFUSED</code> when the python server is offline.
Job Alert Dispatch <code>/api/jobs/alert</code>	FALSE POSITIVE	Returns HTTP 200 indicating successful email dispatch even when <code>RESEND_API_KEY</code> is completely unconfigured.
AI Provider Keys <code>.env.local</code>	GRACEFUL FALLBACK	All LLM keys (Gemini, Groq, OpenAI, Anthropic) are empty; system gracefully routes to internal cognitive mock fallback once authenticated.

2. Deep-Dive: Authentication & Cookie Desynchronization

Root Cause of AI Assistant Failure: The client-side React store (`lib/store.tsx`) operates independently of the Next.js server cookie jar (`cf_session`).

When a user signs in via Google, GitHub, or Phone in `components/auth/AuthGate.tsx`, or when an existing session is rehydrated from `localStorage` on browser refresh:

- The client-side state sets `user = { name, email, authProvider }` in React state and `localStorage`.
- **However**, `/api/auth/login` **is NEVER invoked**. Therefore, the browser never receives a Set-Cookie header with the cryptographically signed HMAC token (`cf_session`).
- In `app/api/assistant/chat/route.ts` (line 168):

```
const authUser = await getAuthenticatedUser();
if (!authUser) {
  return createApiResponse("UNAUTHORIZED", "Authentication required to interact with the assistant.", re
```

- Because the cookie is absent, the backend immediately rejects the request with HTTP 401.
- In `components/assistant/AssistantHome.tsx` (line 1010), the catch block receives status 401 and outputs:

```
⚠ Please sign in to interact with the assistant.
```

The user is left completely confused because the UI shows they are already signed in.

3. Deep-Dive: Voice Assistant & Speech Race Conditions

3.1 Silent Failure on Voice Onboarding

The platform includes an accessible 5-step conversational onboarding flow in

`components/accessibility/GlobalVoiceDictator.tsx`. When an unauthenticated visitor speaks a general question or inquiry, `askAiAssistant()` invokes `/api/assistant/chat`:

```
const res = await fetch("/api/assistant/chat", { method: "POST", body: JSON.stringify(...) });
const data = await res.json();
const replyText = data.reply || ""; // data is { error: { code: "UNAUTHORIZED" } }, replyText becomes ""
```

Because `data.reply` is undefined, `replyText` evaluates to empty string `""`. The voice assistant produces zero speech synthesis output and silently aborts without any audible error chime or feedback.

3.2 Hanging Voice Input (Missing `isFinal` Flag)

In `components/assistant/AssistantHome.tsx`, auto-submission of spoken text relies on a silence countdown:

```
if (isFinal) {
  startSilenceAutoSendCountdown(); // Only starts if the Web Speech API emits isFinal: true
}
```

On many desktop browsers (such as Microsoft Edge or Chrome under ambient background noise), the Web Speech API repeatedly emits interim transcript events without marking `isFinal: true` until long after the user has stopped talking. As a result, the 3.5-second auto-send timer never triggers, causing the spoken message to remain frozen in the input box indefinitely.

3.3 Three Competing Speech Recognition Controllers

The application mounts three overlapping microphone managers:

1. `GlobalVoiceDictator.tsx` (mounted globally in `app/layout.tsx`)
2. `FloatingControlBar.tsx` (mounted globally in `app/layout.tsx`)
3. `AssistantHome.tsx` (mounted on the main assistant route)

When navigating to the assistant page, `GlobalVoiceDictator` attempts to yield control by checking `document.querySelector('[data-assistant-home="true"]')`. However, when toggling mic states via keyboard shortcuts (`Alt+V`) or clicking the floating bar, events collide: one controller attempts to abort while another starts, leaving the UI state out of sync (one indicator shows "listening" while the other shows "idle").

4. Test Suite Decoupling & Design Token Compliance

4.1 Hardcoded Port 8000 Dependency

Running `npm run test:unit` fails on test #3 in `tests/unit/v3_comprehensive_suite.test.mjs`:

```
test("3. Alexa-Style Confirmation Loop ($6): Verification and commit flow", async () => {
  const res1 = await fetch("http://127.0.0.1:8000/api/voice/intent", { ... }); // ECONNREFUSED
});
```

The test suite requires the external Python FastAPI service (`python_ai/server.py`) to be concurrently running. The Next.js unit test suite should mock external service calls or use internal Node.js fallbacks so that CI/CD and offline testing succeed.

4.2 88 Design Token Linter Violations

Running `node scripts/lint_design_tokens.mjs` flags 88 violations of raw hex codes:

- `components/landing/LandingPage.tsx` : `#C7CCD1` , `#06080A` , `#62676D` , `#8B9096`
- `components/ubix/UbixCareerGraph.tsx` : `#7DE1EA` , `#121518` , `#0D0F12` , `#080A0D` , `#181D22` , `#12161A` , `#A5ABB2`
- `components/ubix/UbixHeroScene.tsx` : `#7DE1EA`

The design system specification requires all components to use semantic design tokens (e.g. `bg-bg` , `text-ink` , `bg-surface` , `text-accent`) to ensure consistent dark mode, light mode, and high-contrast accessibility.

5. Comprehensive Remediation Action Plan

#	Area	Prescribed Remediation	Target Files
1	Session Cookie Sync	In <code>lib/store.tsx</code> , update <code>signIn</code> , <code>signInWithGoogle</code> , and <code>signInWithPhone</code> to call <code>/api/auth/login</code> so that the HTTP <code>cf_session</code> cookie is set concurrently with <code>localStorage</code> . Allow automatic guest session provisioning in <code>/api/assistant/chat</code> to eliminate 401 barriers.	lib/store.tsx app/api/assistant/chat/route.ts
2	Voice Auto-Send Fix	In <code>AssistantHome.tsx</code> , start the 3.5s silence auto-send timer on any incoming transcript text (with 1.5s debouncing), rather than strictly requiring <code>isFinal === true</code> .	components/assistant/AssistantHome.tsx
3	Voice Controller Coordination	Centralize the Web Speech instance in a single unified provider (<code>GlobalVoiceProvider</code>) so that <code>FloatingControlBar</code> , <code>GlobalVoiceDictator</code> , and <code>AssistantHome</code> share one authoritative listening state.	lib/voice.ts GlobalVoiceDictator.tsx
4	Tokenize Stray Hex Colors	Refactor the 88 hardcoded hex values in <code>LandingPage.tsx</code> and <code>UbixCareerGraph.tsx</code> to semantic CSS variables and Tailwind classes.	LandingPage.tsx UbixCareerGraph.tsx
5	Decouple Unit Test Suite	Mock or gracefully handle offline responses for <code>http://127.0.0.1:8000</code> in <code>tests/unit/v3_comprehensive_suite.test.mjs</code> so that tests run hermetically.	v3_comprehensive_suite.test.mjs
6	Job Alert Validation	Update <code>/api/jobs/alert</code> to explicitly flag when <code>RESEND_API_KEY</code> is absent (returning a simulation notice or warning) rather than returning an erroneous 200 success.	app/api/jobs/alert/route.ts

6. Conclusion

The core infrastructure of CareerForge / ubix is robust and feature-complete. The failures reported by users are not caused by underlying algorithmic breakdowns, but by **boundary disconnections**: client-server auth desynchronization, unhandled speech state transitions, and missing background service mocks. Executing the targeted remediations above will restore full functionality across the AI Assistant, Voice Engine, and Test Suite.